NCERT Class 10 Physics – Chapter 12: Electricity

CHAPTER 12: ELECTRICITY

12.1 ELECTRIC CURRENT AND CIRCUIT

Electric current is expressed by the amount of charge flowing through a particular area in unit time. If a net charge Q, flows across any cross-section of a conductor in time t, then the current I, through the cross-section is I = Q / t. The SI unit of electric charge is coulomb (C), which is equivalent to the charge contained in nearly 6 x 10^18 electrons. The electric current is expressed by a unit called ampere (A), named after the French scientist, Andre-Marie Ampere.

12.2 ELECTRIC POTENTIAL AND POTENTIAL DIFFERENCE

Electric potential difference between two points in an electric circuit carrying some current is the work done to move a unit charge from one point to the other. Potential difference (V) between two points = Work done (W) / Charge (Q), that is V = W / Q. The SI unit of electric potential difference is volt (V), named after Alessandro Volta. One volt is the potential difference between two points in a current carrying conductor when 1 joule of work is done to move a charge of 1 coulomb from one point to the other.

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12.3 OHM'S LAW AND RESISTANCE

In 1827, a German physicist Georg Simon Ohm found out the relationship between the current I, flowing in a metallic wire and the potential difference across its terminals. The potential difference, V, across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, provided its temperature remains the same. This is called Ohm's law. In other words, V is proportional to I, or V / I = constant = R, where R is a constant for the given metallic wire at a given temperature and is called its resistance. Its SI unit is ohm, represented by the Greek letter Omega.

12.4 RESISTORS IN SERIES AND PARALLEL

When several resistors are joined in series, the total resistance of the combination Rs equals the sum of their individual resistances: Rs = R1 + R2 + R3. The current through each resistor in series is the same, while the total potential difference is equal to the sum of the potential differences across individual resistors. In a parallel combination of resistors, the reciprocal of the equivalent resistance Rp is equal to the sum of the reciprocals of the individual resistances: 1 / Rp = 1 / R1 + 1 / R2 + 1 / R3. The potential difference across each resistor in parallel is identical and equal to the applied voltage.